

PHASE CHANGE MATERIALS (PCM) IN CONSTRUCTION AS A SUSTAINABLE STRATEGY: STATE-OF-THE-ART REVIEW.

Phase Change Materials (PCM) in Construction as a Sustainable Strategy: State-of-the-Art Review.

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Abstract

The construction sector is responsible for a significant portion of global energy consumption and CO₂ emissions. To reduce this impact, various strategies focus on optimizing energy efficiency in buildings and developing innovative materials with thermal energy storage (TES) capacity. In this context, phase change materials (PCM) emerge as a viable alternative to regulate temperature through the absorption and release of latent heat. This research analyzes the technical opportunities and challenges of incorporating PCM into construction materials. The methodology is based on a critical review of three main aspects: classification and comparison of PCM types, integration methods in cementitious matrices, and their sustainable performance. It also explores the combination with construction and demolition waste (CDW) to create more sustainable materials within the circular economy. The results aim to provide a comprehensive view of the potential use of PCM, facilitating the comparison of characteristics and incorporation methods, identifying strategies to optimize thermal and mechanical performance, and promoting integration with conventional and alternative materials such as CDW. Thus, it drives the development of sustainable and energy-efficient construction solutions, reducing the environmental impact of the sector.

Keywords: PCM, thermal energy storage, energy efficiency, CDW

1 Introduction

The construction sector is one of the main contributors to global energy consumption and greenhouse gas (GHG) emissions, playing a central role in the current environmental crisis (UN, 2019). This situation has driven the search for sustainable strategies to reduce dependence on conventional air-conditioning systems, which, although effective, account for nearly 20% of global electricity use and significantly increase CO₂ emissions (IEA, 2022). Thermal Energy Storage (TES) systems are generally divided into two mechanisms: sensible heat storage (SHS) and latent heat storage (LHS). While SHS often relies on recycled materials in construction components, LHS—based on Phase Change Materials (PCM)—has proven more efficient due to its thermophysical properties and ability to moderate thermal fluctuations (Junaid et al., 2024). Passive TES technologies, therefore, have emerged as efficient alternatives for enhancing building performance. Among them, PCM stand out for their capacity to absorb and release latent heat during phase transitions within narrow temperature ranges, passively stabilizing indoor environments (Xianliang et

al., 2022). Several studies have confirmed the effectiveness of PCM in cement-based materials. Djamai et al. (2021), for instance, reported that incorporating 10% of microencapsulated PCM into textile-reinforced concrete slabs reduced peak temperature by up to 4 °C and achieved 37% energy savings without compromising structural integrity. Such findings validate the role of PCM in improving thermal performance within sustainable construction.

The integration of Construction and Demolition Waste (CDW) into PCM systems further reinforces the circular economy approach. Mankel et al. (2019) showed that porous recycled aggregates impregnated with PCM enhanced thermal storage while reducing environmental impact, combining waste valorization with energy efficiency. This synergy provides a dual sustainable strategy for the building sector. Despite their potential, PCM-based technologies still face challenges related to PCM selection, compatibility with cementitious matrices, influence on mechanical properties, and the need for comprehensive assessments of economic and environmental feasibility across the material's life cycle. In light of these aspects, this paper presents a critical state-of-the-art review on the application of PCM in construction materials, with particular emphasis on their combination with recycled aggregates. Opportunities, technical challenges, and emerging trends are examined to provide an integrated perspective on the role of PCM in advancing sustainable, resilient, and energy-efficient buildings.

2 Selection criteria for references

To conduct a critical and representative state-of-the-art review, a multi-phase search strategy was applied across indexed databases (Scopus, Web of Science, ScienceDirect, and Google Scholar), prioritizing articles published between 2014 and 2024. Keyword combinations included phase change materials (PCM), thermal storage, cement-based materials, sustainability, and recycled waste.

Priority was given to systematic reviews, recent experimental studies, and publications in high-impact journals, such as those by Elsevier. The selection was structured around three main thematic axes: (i) classification of PCM (organic, inorganic, and eutectic), (ii) methods of incorporation into construction materials (direct addition, impregnation, encapsulation), and (iii) evaluation of thermal and mechanical performance.

In addition, emerging studies addressing less explored areas were considered, such as PCM durability in recycled matrices or behavior under extreme conditions. The selection process was complemented by a bibliometric network analysis (Figure 1), which identified central articles, interconnected research lines, and evolving topics. Green nodes represent key reviews or seminal works structuring the field, while peripheral nodes reflect exploratory studies that contribute to underdeveloped or future research directions.

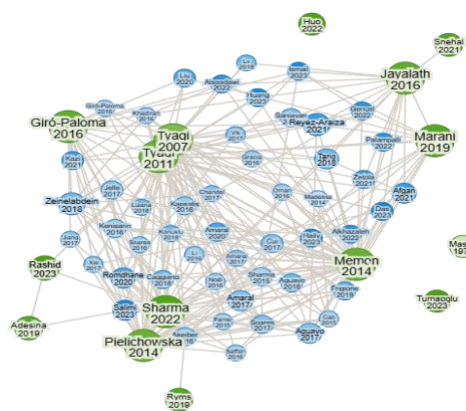


Figure 1 – Bibliographic networks.

3 Phase change materials (PCM)

The integration of Phase Change Materials (PCM) in construction has been consolidated as an effective strategy to enhance building energy efficiency and promote sustainability (Marani et al., 2019). PCM store and release heat through phase transitions, mainly solid–liquid, within comfort temperature ranges, thus contributing to indoor thermal stabilization and reducing HVAC energy demand (Tyagi, 2005). PCM can be classified by composition or phase transition mechanism, aspects that are further detailed in the following subsections together with their functioning, benefits, and technical comparison.

3.1 Functioning and benefits of PCM

PCM store and release thermal energy through latent heat during phase transitions, typically solid–liquid. Throughout this process, the material remains at an almost constant temperature, absorbing or releasing large amounts of heat without abrupt ambient variations (Tyagi, 2005). This property makes them highly efficient passive thermal regulators, reducing indoor temperature peaks and HVAC energy demand (Memon, 2014). Studies have reported up to 20% energy savings in warm climates when PCM are incorporated into walls or roofs (Sharma et al., 2022). Their effectiveness depends on selecting the correct phase change temperature, compatibility with construction materials, and proper architectural integration.

3.1.1 Types of PCM

PCM are typically classified by chemical composition into organic, inorganic, and eutectic groups (Tyagi, 2005; Giro-Paloma et al., 2016). Another approach is based on phase transition mechanism—solid–liquid, solid–solid, solid–gas, and liquid–gas—where solid–liquid transitions are most used in buildings due to their high latent heat in human comfort ranges. Both classifications are complementary, supporting informed selection by considering thermodynamic, chemical, and economic factors (Tyagi, 2005; Giro-Paloma et al., 2016).

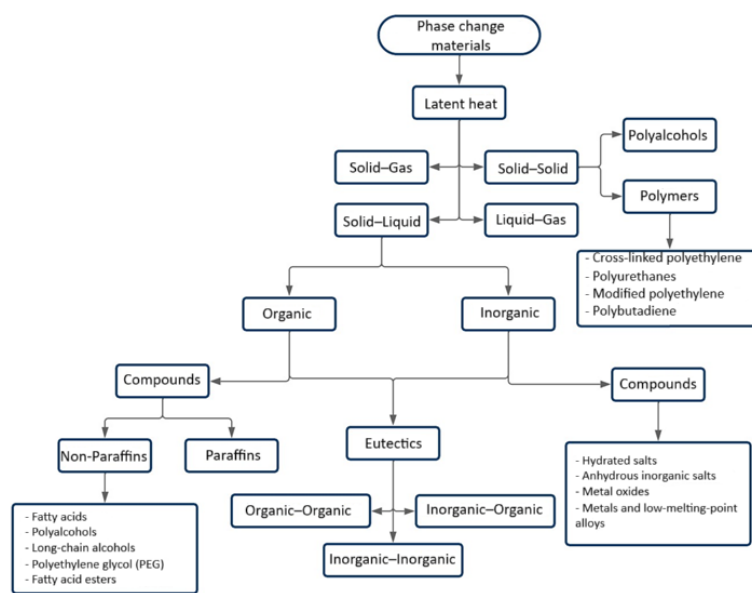


Figure 2 – Classification of PCM

3.1.1.1 Inorganic PCM

Inorganic PCM, such as hydrated salts and metals, are characterized by high energy density and low cost (Sharma et al., 2022). However, issues like corrosion, subcooling, and long-term instability have limited their large-scale adoption. Current research focuses on encapsulation and additives to improve durability and compatibility (Memon, 2014). Despite these challenges, they remain attractive when cost and efficiency are priorities, provided mitigation measures are implemented.

3.1.1.2 Organic PCM

Organic PCM mainly include paraffins and fatty acids, noted for congruent melting, chemical stability, and low corrosivity (Tyagi, 2005; Giro-Paloma et al., 2016). They offer good thermal reliability and a wide range of melting points, making them versatile for different TES applications. Their main drawbacks are low thermal conductivity and, in the case of some paraffins, flammability. Nevertheless, their stability under repeated cycles and ease of integration make them a preferred choice for passive thermal storage solutions.

3.1.1.3 Eutectic mixtures

Eutectic PCM are formed by combining two or more components that melt or solidify congruently at fixed ratios, yielding constant and tunable melting points (Tyagi, 2005; Giro-Paloma et al., 2016). They can be organic, inorganic, or hybrid, which broadens their operational temperature range. Advantages include low subcooling and good thermal stability, although challenges remain in long-term characterization and limited commercial availability. Their adaptability positions them as a promising research line for tailored thermal solutions in construction.

4 Incorporation methods

The selection of the incorporation method for PCM into cementitious matrices is decisive for the success of thermal storage solutions in buildings. Figure 3 illustrates the main alternatives and highlights that, in practice, the choice can be adapted or combined

according to thermal, structural, and economic requirements. Direct incorporation is attractive for its simplicity and low cost, but long-term viability is limited by compatibility and durability issues. Impregnation into porous materials improves retention and stability, though capacity depends on porosity and support type. Encapsulation, especially at micro and nano scales, offers the highest efficiency and control, but faces challenges in cost and industrial scalability. Current trends point toward hybrid methods and optimization of existing techniques, with real-scale validation and long-term durability studies being essential for widespread adoption.

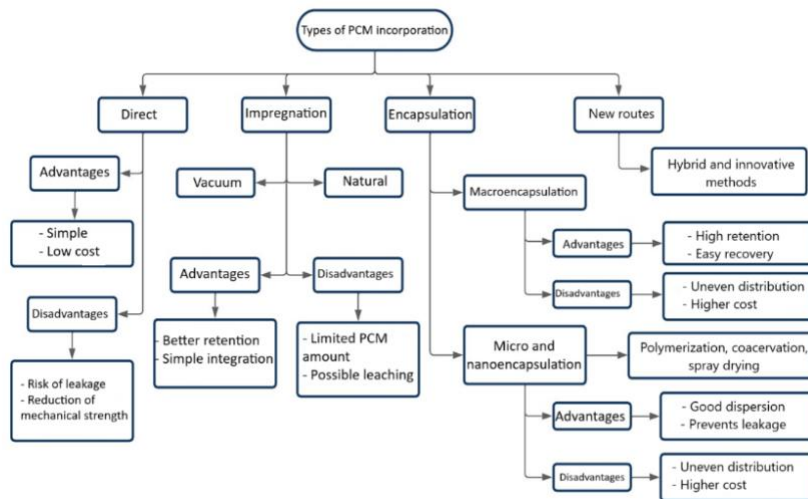


Figure 3 – PCM incorporation methods

4.1 Direct incorporation

Direct incorporation into cementitious matrices is the simplest and cheapest method since it requires no encapsulation or additional treatments (Rodríguez et al., 2021). However, it presents serious limitations in structural applications. Some organic PCM are incompatible with the alkaline environment of cement, leading to chemical reactions that weaken the microstructure and reduce compressive strength. Moreover, without confinement, PCM migration during thermal cycles compromises efficiency and durability. Despite these drawbacks, direct incorporation remains useful in preliminary research and for developing new composites, provided mitigation measures—such as additives, surface treatments, and optimized mix designs—are applied.

4.2 Impregnation into porous materials

This method introduces PCM into porous supports or lightweight aggregates such as perlite, expanded vermiculite, calcined clay, pumice, recycled aggregates, and increasingly, construction and demolition waste (CDW) (Marani & Nehdi, 2019; Junaid et al., 2024). PCM penetrates the porous structure through natural absorption or vacuum techniques, and the impregnated support is later incorporated into concrete or mortar.

This approach improves compatibility by avoiding direct contact between PCM and cementitious environments, reducing interference in setting and enhancing thermal stability across cycles (Palacios et al., 2023). PCM confinement within the support also minimizes leakage. The use of CDW as support aligns with circular economy principles, as recycled aggregates show absorption capacities comparable to traditional lightweight aggregates

(Junaid et al., 2024). Limitations include storage capacity restricted by porosity and potential leakage without additional coatings. Key parameters such as pore size and total porosity determine long-term performance (Flórez Rodríguez, 2021).

Common PCM: medium-chain paraffins (C16–C20), fatty acids such as lauric and palmitic. Frequent supports: expanded perlite, vermiculite, porous sand, lightweight recycled aggregates, CDW.

Applications: thermal mortars, masonry blocks, lightweight prefabricated elements.

4.3 PCM encapsulation

Encapsulation, in its different scales, is the most advanced strategy to ensure PCM stability and functionality in cementitious matrices. However, its widespread adoption depends on synthesis optimization, cost reduction, and long-term durability validation under real service conditions. Macroencapsulation is simple and allows easier maintenance but has limited thermal efficiency due to low conductivity and mechanical fragility. Micro- and nanoencapsulation, although more costly and complex, enable homogeneous and efficient integration, opening the door to innovative applications in buildings (Palacios et al., 2023). Emerging research explores sustainable and multifunctional encapsulating materials, as well as hybrid approaches, to overcome current limitations.

4.3.1 Macroencapsulation

Although macroencapsulation is simple and allows modular maintenance, its thermal efficiency is limited by the low conductivity of the encapsulated material and by the difficulty of ensuring effective thermal contact with the cementitious matrix. Durability also depends on the resistance of the encapsulating materials against thermal cycles, mechanical loads, and corrosive agents. These challenges may restrict service life and increase maintenance costs. In practice, PCM replacement or recharging can be more complex than expected, especially in structural elements. For these reasons, macroencapsulation is recommended for applications where accessibility and maintenance are feasible, with current trends focusing on modular prefabrication and improved encapsulating materials (Palacios et al., 2023).

4.3.2 Microencapsulation

Microencapsulation is one of the most promising strategies for integrating PCM into cementitious matrices, allowing precise thermal release control and minimizing negative impacts on mechanical properties. However, its large-scale viability depends on optimized synthesis, cost reduction, and long-term durability validation. Capsule rupture during mixing or service may compromise efficiency, highlighting the importance of resistant and compatible encapsulating materials. Current trends explore multifunctional microcapsules that provide additional properties such as self-healing or fire resistance, broadening the scope of sustainable applications (Palacios et al., 2023).

4.3.3 Nanoencapsulation

Nanoencapsulation represents the technological frontier for PCM integration, offering unique advantages in thermal efficiency, dispersion, and mechanical reinforcement. However, widespread use is hindered by high synthesis costs, specialized equipment needs, and

insufficient long-term validation. Concerns remain regarding nanoparticle release during use or recycling, raising environmental and regulatory questions. Research is currently oriented toward sustainable and multifunctional nanocapsules and interdisciplinary collaboration to overcome these challenges (Palacios et al., 2023).

5 Technical comparison of PCM types

The technical comparison of PCM confirms that no universal option exists; selection must consider thermal performance, matrix compatibility, incorporation method, and durability requirements. Organic PCM (paraffins, fatty acids) dominate due to stability, wide compatibility, and latent heats >200 kJ/kg, making them suitable for passive storage (Sharma et al., 2022; Memon, 2014; Marani et al., 2019). Yet, their low conductivity (~0.2 W/m·K) and paraffin flammability reduce efficiency and impose safety constraints, while fatty acids may cause odor and corrosion (Tyagi, 2005). Inorganic PCM provide higher energy density and conductivity but require robust encapsulants to prevent corrosion and segregation, whereas eutectic mixtures offer design flexibility but face availability and characterization barriers. Current trends emphasize encapsulation, conductive additives, recycled supports, and full-scale validation to ensure durability and feasibility (Sharma et al., 2022; Memon, 2014; Marani et al., 2019).

Table 1 – Melting point and latent heat of selected organic PCM (adapted from Sharma et al., 2022; Memon, 2014).

Organic PCM	Melting point (°C)	Latent heat (kJ/kg)	Organic PCM	Melting point (°C)	Latent heat (kJ/kg)
1-Dodecanol	26	200	Hexadecane	18.1	236
Capric acid	30.1	158	Methyl palmitate	29	205
Lactic acid	26	184	Octadecane	28–28.1	244–250.7
Butyl stearate	19	140	Octadecyl 3-mercaptopropylate	21	143
Butyl stearate	19	140	Octadecyl thioglycolate	26	90
Camphenilone	39	205	Paraffin C17	21.7	213
Caprylone	40	259	Paraffin C18	28	244
Dimethyl sebacate	21	120–135	Paraffin wax RT45	45	160
Docasyl bromide	40	201	Paraffin wax RT55	55	170
Emerest 2325	20	134	Paraffin C13–C24	22–24	189
Emerest 2326	20	139	Polyglycol E600	22	127.2
Formic acid	7.8	247	Propyl palmitate	19	186
Glycerin	17.9	198.7	Tetradecane (C14H20)	5	172.21
Glycerin	17.9	198.7	Vinyl stearate	27–29	122
Heptadecane	20.8–21.7	171–172	–	–	–

In contrast, inorganic PCM, mainly hydrated salts, are characterized by higher energy density and superior thermal conductivity (0.5–0.7 W/m·K) compared to organics, along with generally lower cost (see Table 2). These characteristics make them attractive for applications requiring efficient heat transfer and greater storage capacity in smaller volumes (Memon, 2014; Tyagi, 2005; Marani et al., 2019). However, their use faces technical challenges: corrosivity may compromise encapsulants and cement matrices, supercooling hinders controlled energy release, and phase segregation can reduce cyclic stability and system durability. Mitigation strategies include resistant encapsulants, nucleating additives, and hybrid formulations.

Metals and alloys, although offering even higher thermal conductivities ($>1.0 \text{ W/m}\cdot\text{K}$), are restricted by high cost and weight, limiting their use to specialized applications where efficiency justifies the expense (Tyagi, 2005). Eutectic mixtures, both organic and inorganic, allow precise melting-point adjustment and usually exhibit good thermal stability with low supercooling, making them flexible alternatives for specific thermal requirements (see Table 2). However, large-scale characterization and commercial availability remain limited. Their main advantage lies in tailoring materials to the desired operating temperature, which represents a promising research path with high technological potential for innovative building applications (Sharma et al., 2022; Memon, 2014; Tyagi, 2005).

Table 2 – Melting point and latent heat of selected inorganic and eutectic PCM (adapted from Sharma et al., 2022; Memon, 2014).

Inorganic			Eutectic		
PCM	Melting point (°C)	Latent heat (kJ/kg)	PCM	Melting point (°C)	Latent heat (kJ/kg)
$\text{CaBr}_2 \cdot 6\text{H}_2\text{O}$	34	138	$14\% \text{ LiNO}_3 \cdot 3\text{H}_2\text{O} + 86\% \text{ Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	72	180
$\text{CaCl}_2 \cdot 12\text{H}_2\text{O}$	29.8	174	$34\% \text{ Myristic acid } (\text{C}_{14}\text{H}_{28}\text{O}_2) + 66\% \text{ Capric acid } (\text{C}_{10}\text{H}_{20}\text{O}_2)$	24	147.7
$\text{FeBr}_3 \cdot 6\text{H}_2\text{O}$	27	105	$47\% \text{ Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O} + 53\% \text{ Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	30	136
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	37	223	$50\% \text{ Acetamide } (\text{CH}_3\text{CONH}_2) + 50\% \text{ Urea } (\text{NH}_2\text{CONH}_2)$	27	163
$\text{K}_2\text{HPO}_4 \cdot 6\text{H}_2\text{O}$	14	109	$50\% \text{ CaCl}_2 + 50\% \text{ MgCl}_2 \cdot 6\text{H}_2\text{O}$	25	95
$\text{KF} \cdot 4\text{H}_2\text{O}$	19	231	$53\% \text{ Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} + 47\% \text{ Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	66	168
$\text{LiBr}_2 \cdot 2\text{H}_2\text{O}$	34	124	$55\% \text{ CaCl}_2 \cdot 6\text{H}_2\text{O} + 45\% \text{ CaBr}_2 \cdot 6\text{H}_2\text{O}$	14.7	140
$\text{LiClO}_3 \cdot 3\text{H}_2\text{O}$	8	253	$60\% \text{ Sodium acetate} \cdot 3\text{H}_2\text{O} + 40\% \text{ Urea}$	30–31.5	200.5–226
$\text{LiNO}_3 \cdot 2\text{H}_2\text{O}$	30	296	Capric/Lauric/Palmitic acid	20.75	134
$\text{LiNO}_3 \cdot 3\text{H}_2\text{O}$	30	296	Capric/Myristic/Palmitic acid	17	131.7
$\text{Mn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	25.8	125.9	Capric/Palmitic/Stearic acid	19.83	154.11
$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$	32–36	246.5–247	Gallium (Ga)	30	80.9
$\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$	35–36	265–281	Hexadecanol/Palmitic/Lauric acid	26.53	179.63
$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$	31–32.4	251.1–254	Lauric/Palmitic/Stearic acid	36.79	159
$\text{Na}_2\text{SO}_4 \cdot 3\text{H}_2\text{O}$	32	251	Octadecane + Docosane	25.5–27	203.8
$\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	36–36.4	146.9–147	Octadecane + Heneicosane	25.8–26	173.93
			Oleic acid/Isopropyl palmitate/Butyl stearate	5.14	104.12

Figure 4, represented through a radar chart, enables visual comparison of the strengths and weaknesses of the main PCM types in terms of thermal and operational properties. It is evident that no material gathers all the ideal characteristics for construction applications, since each type presents trade-offs such as low thermal conductivity, leakage risk, or cyclic stability limitations. However, these weaknesses should not be seen as absolute constraints. Incorporation techniques and formulations can compensate for many of them; for example, the leakage tendency of organic PCM can be mitigated by microencapsulation, while their low conductivity can be improved with conductive fillers. Hence, PCM selection should rely on a holistic analysis of pros and cons, considering intrinsic properties, incorporation techniques, operating environment, and durability requirements (Memon, 2014; Marani et al., 2019).

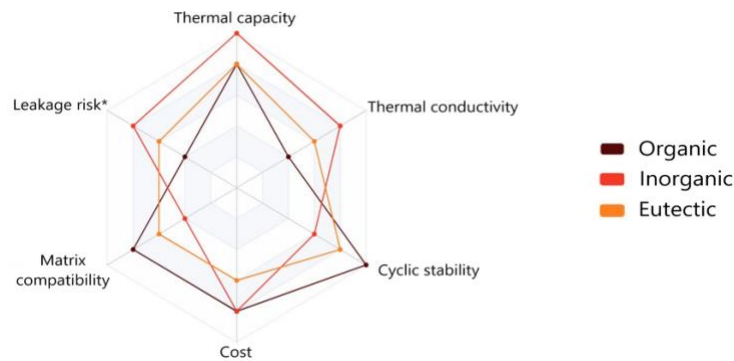


Figure 4 – Comparison of PCM types according to thermal and operational properties..

Complementarily, Figure 5 presents a decision diagram guiding the selection of both PCM type and incorporation method, integrating technical, economic, and application-related criteria. This tool is especially useful in engineering and building projects as it structures the key variables in the selection process. Thus, not only the fundamental material properties are considered, but also cost, incorporation feasibility, and environmental conditions. In this way, selection becomes a multidimensional evaluation that promotes tailored solutions for specific project needs.

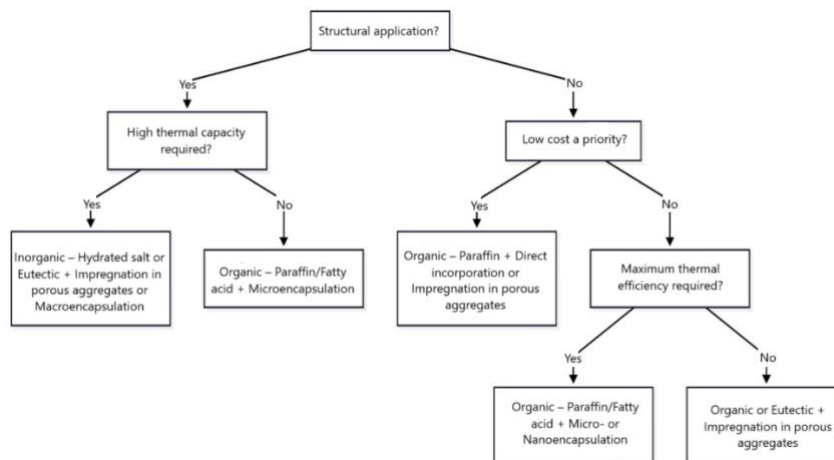


Figure 5 – Decision diagram for PCM and incorporation method selection.

6 Recent studies and applications of PCM

The evolution of the use of phase change materials (PCM) in construction can be traced through a series of review articles that have shaped the field and guided subsequent experimental research.

1. **Foundational review (2005).** The state-of-the-art review by Tyagi et al. (2005) was one of the first comprehensive efforts to catalog thermal storage in buildings, compiling nearly 150 experimental studies published to that date. The paper emphasized the need to combine thermal capacity with chemical stability and identified early challenges such as PCM leakage and the lack of unified testing standards.
2. **Thematic consolidation (2012–2014).** The compendium by Cabeza et al. (2012) and the review by Memon (2014) broadened the focus to PCM integration in wall

elements and concrete. Memon (2014) analyzed more than 250 studies, highlighting the transition from direct mixing and impregnation techniques toward early experiments with macro- and micro-encapsulation. Challenges identified included alkaline compatibility and long-term mechanical characterization.

3. **Specialization in microencapsulation (2016).** The review by Giro-Paloma et al. (2016) delved into microencapsulation, analyzing 449 publications on MPCM between 2005 and 2015. This work systematized synthesis methods, core and shell materials, as well as characterization protocols, marking the transition from simple PCM inclusion to optimization of dispersion and internal stability.
4. **Diversification of methods and critical focus (2019).** Marani and Nehdi (2019) included more than 120 recent studies and compared four main incorporation methods: microencapsulation, macroencapsulation, porous aggregates, and shape-stabilized composites. Their work emphasized the need to mitigate compressive strength loss and improve durability under thermal cycling.
5. **Sustainability and real-condition validation (2022–2024).** Sharma et al. (2022) and Barbhuiya et al. (2024) compiled more than 300 experimental references, integrating for the first time circular economy principles, life cycle assessment, and emerging standards. Recent reviews highlight the exploration of hydrated salts, tailor-made eutectic mixtures, and hybrid composites with conductive additives or recycled materials.

Table 3 highlights the diversification of PCM types, incorporation techniques, and applications, but reported results reveal inconsistencies. Mechanical strength reductions vary widely—from minor losses to >50%—depending on chemical nature, method, and dosage. Microencapsulation reduces leakage but weakens interfaces, while macroencapsulation preserves integrity but risks poor distribution. Organics show better stability and compatibility yet suffer from low conductivity and leakage, whereas inorganics, particularly hydrated salts, offer higher capacity but face supercooling, corrosion, and strength issues. Application context is crucial: small losses may be acceptable in thermal storage systems but not in structural elements. Current trends emphasize hybrid PCM, recycled supports, and tailored mixtures, though clear standards on acceptable strength reduction and long-term protocols remain lacking.

Table 3 – Recent studies on PCM use in construction: type, incorporation, application, and key results (adapted and compiled from various authors, 2016–2024).

Author & year	PCM	Incorporation type	Application	Key results
D'Alessandro et al. (2018)	Paraffin	Microencapsulation (MPCM) and macroencapsulation	Thermal storage in buildings	Reduced concrete density ($\approx 2\times$ PCM weight). Compressive strength decreased, but higher structural reliability.
Cao et al. (2018)	Microencapsulated (MPCM) in geopolymer	Direct mixing (5.2% wt.)	Thermal walls	35% less electricity consumption to maintain 23 °C indoors.
Qian & Li (2018)	n-Octadecane/diatomite	Direct mixing (30%)	Thermoregulatory mortar	Stable mechanical strength (3.5 MPa flexural, 18.3 MPa compressive) after 400 thermal cycles.

Tian et al. (2019)	Paraffin RT28 in AAC	Porous impregnation	Thermal insulation	Thermal conductivity increases with paraffin, reducing insulation. Optimal content: 2/3 saturation.
Afgan et al. (2019)	Fatty acids (ternary)	Macroencapsulation in aggregates	Structural–thermal storage	Strength >15 MPa. Reduced internal temperature (6.4 °C) and peak loads (22–30.2 °C).
Ramakrishnan et al. (2021)	Paraffin/hydrophobic expanded perlite	Direct mixing (15–30%)	Thermoregulatory geopolymer foam	Peak indoor temperature reduction: 1.85 °C (15% PCM) and 3.76 °C (30% PCM). Thermal capacity +105181%.
Kalombe et al. (2023)	Paraffin/soybean oil blend	Latex coating	Anti-freezing protection	Higher heat storage and shorter freezing time. Reduced need for de-icing salts.
Liua et al. (2018)	Hydrated salts (EHS/EGO, EHS/P(AA-AA))	Direct mixing (5–20% wt.)	Cement mortar	Reduced mechanical strength but viable for envelopes. Improved thermal stability.
Mohseni et al. (2020)	Inorganic PCM (macroporous ceramic)	Double-layer macroencapsulation	Lightweight aggregates (TESA)	Compressive strength reduction 6–9%. Thermal stability after cycles (15–40 °C). Cooler surface vs. control.
Martelletto et al. (2022)	Salts (40% KNO ₃ –60% NaNO ₃) in diatomite	Direct mixing (5% wt.)	High-temperature storage (400 °C)	40% integration doubles stored/released energy.
Wang et al. (2018)	Fatty acids (lauric, myristic, palmitic)	Impregnation in diatomite/ceramsite	Storage aggregates	Strength >18 MPa with 80% aggregates. Better hydration temperature control vs. ceramsite.
Cui et al. (2022)	Paraffin	Macroencapsulation in steel spheres (PCM-HSB)	Energy piles	Steel fiber reinforcement: +71% thermal conductivity. Greater heat capacity and structural support.
Al-Yasiri & Szabó (2023)	Paraffin	Layer/capsules in bricks	Hot climates (Iraq)	Brick temperature reduction: 5 °C (PCM layer), 30–60 min delay. Layer: better thermal; capsules: better mechanical.
Giro-Paloma et al. (2016)	Organics (paraffins, fatty acids); Inorganics (salts)	Microencapsulation	MPCM applications review	Critical characterization for efficiency. Techniques: thermal analysis, microscopy.
Zhang et al. (2019)	Na ₂ SO ₄ ·10H ₂ O	Microencapsulation in SiO ₂	Thermal storage	Stability after 100 cycles (fusion enthalpy: 100.9 kJ/kg). Melting: 36.2 °C (ΔH=125.6 kJ/kg).
Frigione et al. (2019)	Organics/Inorganics	Micro/Macroencapsulation (3–22%)	Thermoregulatory mortars	Improved thermal efficiency of buildings. Reduced mechanical strength with %PCM.
Das et al. (2023)	Organics/Inorganics	Micro/Macroencapsulation, impregnation (5–20%)	Cementitious materials	Impact on strength, thermal conductivity, and storage.
Belete et al. (2024)	Hydrated salts/graphite	Macroencapsulation (3–10%)	Thermoregulatory mortar	Lower thermal conductivity, higher heat capacity. Slight mechanical reduction. Suitable for construction.

Zakaria et al. (2023)	Hydrated salts	Microencapsulation in PMMA (5%)	Cement mortar	Improved cyclic stability and thermal storage. Slight mechanical strength reduction.
Muhammad et al. (2024)	Hydrated sodium carbonate	Microencapsulation (5–15%)	Mortar/concrete	Improved thermal regulation and storage. Reduced temperature fluctuations. Slight compressive strength decrease.
Farhan et al. (2023)	Salt hydrates	Various (5–20%)	Concrete panels	Reduced indoor temperature, energy savings. Slight mechanical strength reduction.

6.1 Current panorama of PCM use in construction

The review of recent studies —summarized in Table 3— shows that organic PCM, mainly paraffins and fatty acids, remain dominant in cement-based matrices due to stability and compatibility, while hydrated salts gain attention for higher energy density but face challenges such as supercooling, corrosion, and poor alkaline affinity (Marani & Nehdi, 2019; Sharma et al., 2022). Microencapsulation ensures dispersion and minimizes leakage, albeit at higher cost and with reduced conductivity (Tyagi et al., 2010). Macroencapsulation and porous impregnation provide scalable, lower-cost alternatives but require careful structural design (Ryms et al., 2019).

Experimental data confirm that 5–30 wt% PCM can lower peak temperatures up to 6 °C, with strength losses typically <10%, acceptable in non-structural applications (Snehal & Das, 2019). Recent research also explores conductive additives and multifunctional encapsulants to mitigate low conductivity and corrosion (Belete et al., 2024).

Key challenges include: (a) increasing conductivity of organics without losing stability, (b) overcoming supercooling in hydrated salts, and (c) extending service life under thermal and mechanical cycling. Research trends point to hybrid/eutectic composites, circular-economy supports, and full-scale validation with specific standards (Sharma et al., 2022; Barbhuiya et al., 2024). Looking forward, progress in encapsulation, customization, and sustainable integration could expand applications in nearly zero-energy and smart buildings, though stronger applied research and harmonized regulations remain essential.

6.1.1 PCM and methods according to climate and application

Experience shows that there is no universally superior PCM or incorporation method. The key lies in aligning the material’s phase-transition range, the integration technique, and the climatic and functional conditions of the building system. For example, microencapsulated paraffins have proven especially effective in warm climates and lightweight elements, thanks to their melting point being compatible with human comfort and the leakage reduction offered by encapsulation. For applications requiring higher storage capacity, such as massive walls or structural elements in temperate climates, fatty acids and hydrated salts —especially in porous aggregates or macroencapsulation— are more suitable due to their high enthalpy and ability to buffer thermal peaks. In freeze-prone areas, paraffin–vegetable oil blends in flexible coatings delay ice formation and reduce de-icing salt use. For medium- and high-temperature storage, eutectic salts in porous supports allow stable operation at elevated ranges, doubling stored energy compared to conventional solutions. This diversity confirms that selection must always be contextual and based on integral use-condition analysis.

6.1.2 Urgent research gaps

One of the most critical gaps is the lack of long-term durability studies under real service conditions. Most studies report results after a limited number of thermal cycles, representing only a fraction of expected building lifetimes. Moreover, interactions between thermal performance and mechanical properties under real loads, such as fatigue, creep, or microcracking, remain largely unexplored, particularly in structural or large prefabricated elements. Another gap is the absence of multi-scale testing and modeling protocols to enable comparability between laboratories and predict building-scale behavior. Methodological dispersion hinders knowledge accumulation and practical transfer. These gaps slow innovation and create uncertainty for industry and regulators.

6.1.3 From research to practice: barriers and needs

Bridging the laboratory-to-construction gap remains the greatest challenge. The lack of PCM-specific regulations and standards undermines designer confidence and complicates product certification. Furthermore, the absence of open and comparable databases on thermophysical properties, environmental impacts, and life cycle costs prevents informed decisions and objective comparison between alternatives. The scarcity of pilot studies and demonstrators under real, long-term conditions reinforces the gap between research and practice. Finally, there is a lack of comprehensive economic analyses that consider not only PCM cost, but also encapsulation, implementation logistics, and recyclability at end-of-life. Without these elements, PCM adoption in construction will remain marginal and experimental.

6.1.4 Techniques and materials with greatest potential

Among the most promising strategies are conductive additives such as graphite or graphene, which significantly increase thermal conductivity and act as nucleating agents to reduce supercooling in hydrated salts. The use of porous aggregates such as expanded perlite and alternative matrices like geopolymers allows PCM integration into a wider range of building systems while reducing carbon footprint and improving chemical resistance. Reinforcement with metallic or carbon fibers not only improves toughness and conductivity but also enables hybrid structural elements, such as energy piles or prefabricated PCM panels. These strategies simultaneously address the two main PCM limitations —low thermal conductivity and strength loss— opening the way for more robust, sustainable, and context-adaptable solutions.

7 Conclusions

The incorporation of phase change materials (PCM) into construction systems stands out as a promising strategy to tackle energy and environmental challenges in the building sector. Experimental evidence confirms their capacity to reduce energy consumption and enhance thermal comfort, though most studies remain limited to controlled conditions (Marani et al., 2019; Snehal & Das, 2019). Progress toward standardized and viable applications requires validation under real conditions, life-cycle assessments, and the integration of sustainability and circular economy principles (Sharma et al., 2022).

In South America, PCM potential is especially relevant due to climatic diversity and the need for efficient solutions in extreme conditions. Microencapsulated paraffins are suitable for lightweight envelopes and passive cooling, while hydrated salts, though affordable and

energy-dense, still face challenges like supercooling and corrosion (Florez Rodríguez, 2021). For the Latin American market, macroencapsulation, porous aggregates, and flexible coatings offer cost-effective alternatives, complemented by geopolymers and conductive additives that enhance efficiency and durability.

However, barriers such as the absence of specific standards, scarce long-term data, and limited pilot projects persist (Barbhuiya et al., 2024). Strengthening collaboration among academia, industry, and regulators, along with open-access databases, is crucial to accelerate adoption. Ultimately, PCM integration in South American construction not only improves thermal efficiency but also contributes to broader goals of sustainability, resilience, and climate change adaptation.

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